```
---
title: "Sales Dashboard"
output:
  flexdashboard::flex_dashboard:
    orientation: rows
    vertical_layout: fill
    runtime: shiny
    sidebar:
      title: "Filters"
runtime: shiny
---

```{r setup, include=FALSE}
library(flexdashboard)
library(DT)
library(ggplot2)
library(dplyr)
library(shiny)

################
## Dataset
###############
# Sample data (same as flexdashboard)
dates <- seq.Date(as.Date("2024-01-01"), as.Date("2024-12-31"), by = "day")

sales_data <- data.frame(
  date = dates,
  category = sample(c("Electronics", "Clothing", "Food", "Books"), length(dates), replace =
TRUE),
  region = sample(c("North", "South", "East", "West"), length(dates), replace = TRUE),
  sales = runif(length(dates), 100, 1000),
  profit = runif(length(dates), 10, 200)
)
```
```

````
sales_data$customers <- round(sales_data$sales / runif(nrow(sales_data), 10, 50))

# Create a reactive expression for filtered data
filtered_data <- reactive({
  if (input$region == "All") {
    sales_data
  } else {
    sales_data[sales_data$region == input$region, ]
  }
})
```

Sidebar {.sidebar}
=====================================

```{r}
selectInput("region", "Select Region:",
            choices = c("All", unique(sales_data$region)),
            selected = "All")
```

Dashboard
=====================================

Row {.value-boxes}
-------------------------------------

### Total Sales
```{r}
renderValueBox({
  data <- filtered_data()
  valueBox(
````

````
    value = paste0("$", format(round(sum(data$sales), 0), big.mark = ",")),
    icon = "fa-dollar-sign",
    color = "primary"
  )
})
```

### Total Profit
```{r}
renderValueBox({
  data <- filtered_data()
  profit <- sum(data$profit)
  valueBox(
    value = paste0("$", format(round(profit, 0), big.mark = ",")),
    icon = "fa-chart-line",
    color = "success"
  )
})
```

### Total Customers
```{r}
renderValueBox({
  data <- filtered_data()
  customers <- sum(data$customers)
  valueBox(
    value = format(round(customers, 0), big.mark = ","),
    icon = "fa-users",
    color = "info"
  )
})
```
````
````

```
Row
-------------------------------------

### Sales by Category
```{r}
renderPlot({
  data <- filtered_data()
  cat_data <- data %>%
    group_by(category) %>%
    summarise(Total_Sales = sum(sales))

  ggplot(cat_data, aes(x = category, y = Total_Sales, fill = category)) +
    geom_bar(stat = "identity") +
    labs(x = "", y = "Sales ($)") +
    ggtitle("Sales by Category") +
    theme_minimal() +
    theme(legend.position = "none") +
    scale_fill_brewer(palette = "Set2") +
    scale_y_continuous(labels = scales::comma)
})
```

Recent Transactions
```{r}
renderDT({
  data <- filtered_data()
  datatable(
    data,
    options = list(
      pageLength = 5,
      dom = 'Bfrtip',
```
```

```
      buttons = c('copy', 'csv', 'excel'),
      columnDefs = list(
        list(className = 'dt-center', targets = '_all')
      )
    ),
    rownames = FALSE,
    class = 'display compact',
    filter = 'top'
  ) %>%
    formatCurrency(columns = 'sales', currency = "$", interval = 3, mark = ",") %>%
    formatCurrency(columns = 'profit', currency = "$", interval = 3, mark = ",") %>%
    formatDate(columns = 'date', method = 'toLocaleDateString')
})
```